

# Simple Shell Design Document

## 1. Contributions

Esha - handle\_sigint(), add\_history(), print\_history(), print\_report(), main()

Aryaman - parse\_args(), parse\_pipeline(), launch()

## 2. Implementation

### Steps to run on terminal:

1. Path of the file
  2. Make
  3. ./simple-shell
  4. Enter space separated commands only or | for pipes
  5. "history" to display history of the commands
  6. Ctrl-c and then enter for exit and termination report
- When you run, a prompt ourshell> is displayed
  - Command execution using fork and execvp
  - Array of structures (struct Entry\_in\_History to store command history
  - Each entry has - command string, PID of last process in pipeline, start time(time\_t) and duration.
  - add\_history() to record new entries, print\_history() to print the commands, print\_report() for termination report.
  - SIGINT (Ctrl-c) is detected using signal () by flag volatile sig\_atomic\_t sigint\_received. (Referred to gpt to understand and implement)
  - launch() creates pipes, forks child processes, dup2() used to connect I/O and execvp() used.
  - For time tracking, time(NULL) and CLOCK\_MONOTONIC used. (Referred gpt to understand syntax and functioning)
  - We have referred to gpt to understand some library functions that needed to be implemented for time tracking and signals.
  - Except the commands already mentioned in the instructions, our shell also runs commands like date, whoami

## 3. Limitations

- cd command not supported - directory change is required and we have not implemented that (as we can only run the commands in /usr/bin as given in the instructions)
- Background processes not supported - as that would require proper block/unblock management (bg, fg, & not supported)

- \$, “”, /, <, > not supported as we have done splitting of input command as per instructions os assignment.
- The commands input and history entries have a limit.

#### **4. Error Handling**

- Proper failure error handling done for fork, pipe, execvp, waitpid and clock\_gettime as well as for ctrl-c and history memory full.

#### **5. Github repository link: <https://github.com/antelope0112/SIMPLE-SHELL-GROUP66>**